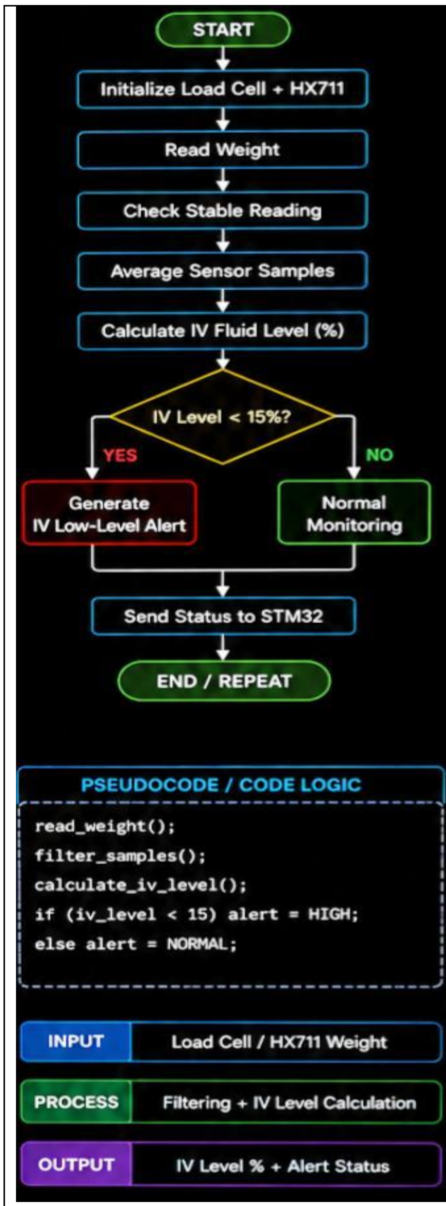
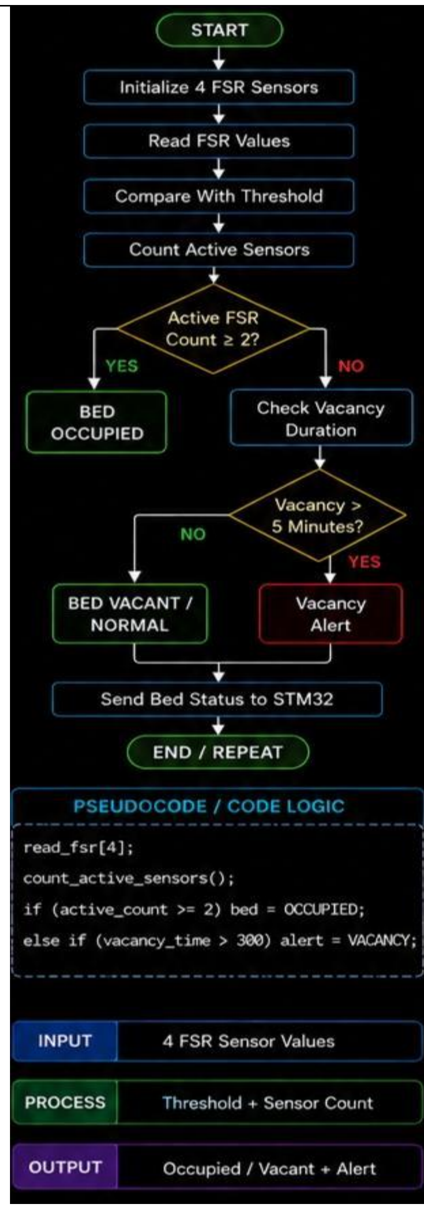
SOFTWARE DESIGN:

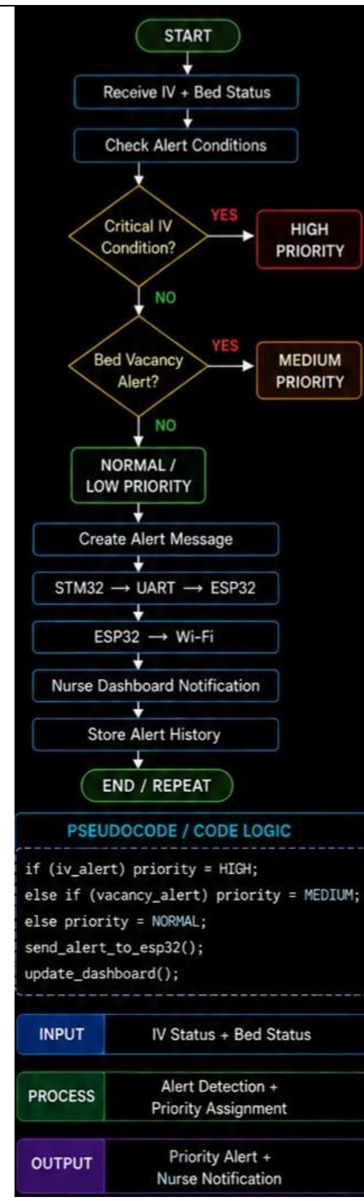
1.FLOW CHART:

2.ALGORITHM :

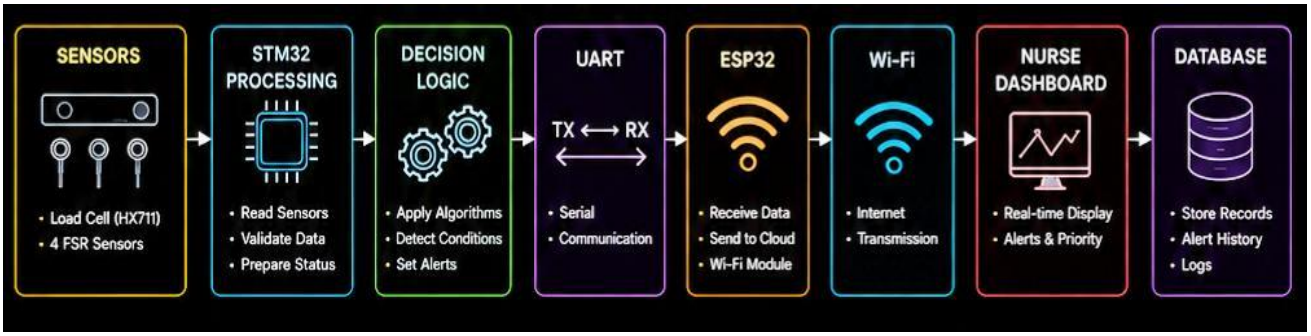
IV FLUID LEVEL MONITORING	BED OCCUPANCY DETECTION	ALERT PRIORITY AND NOTIFICATION





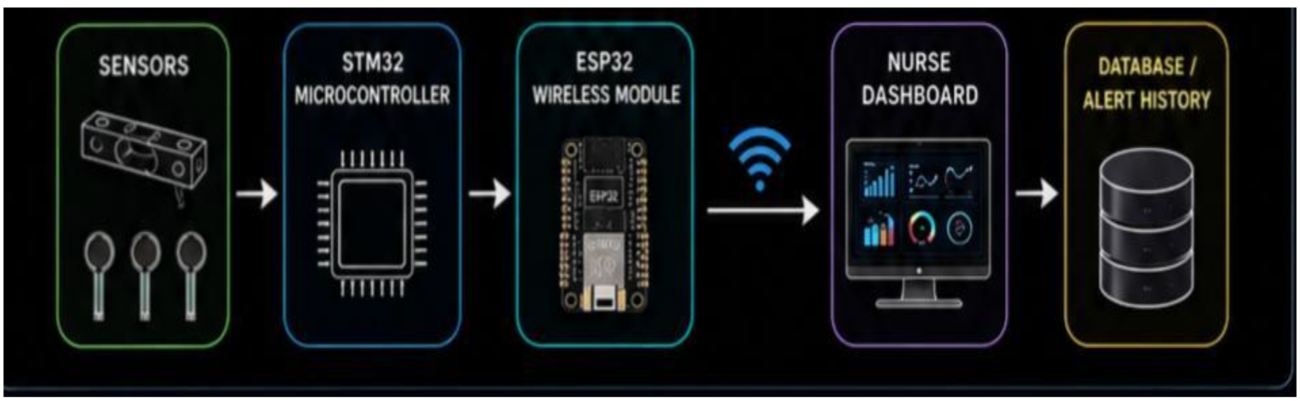


SOFTWARE PIPELINE:



3.COMMUNICATION PROTOCOL:

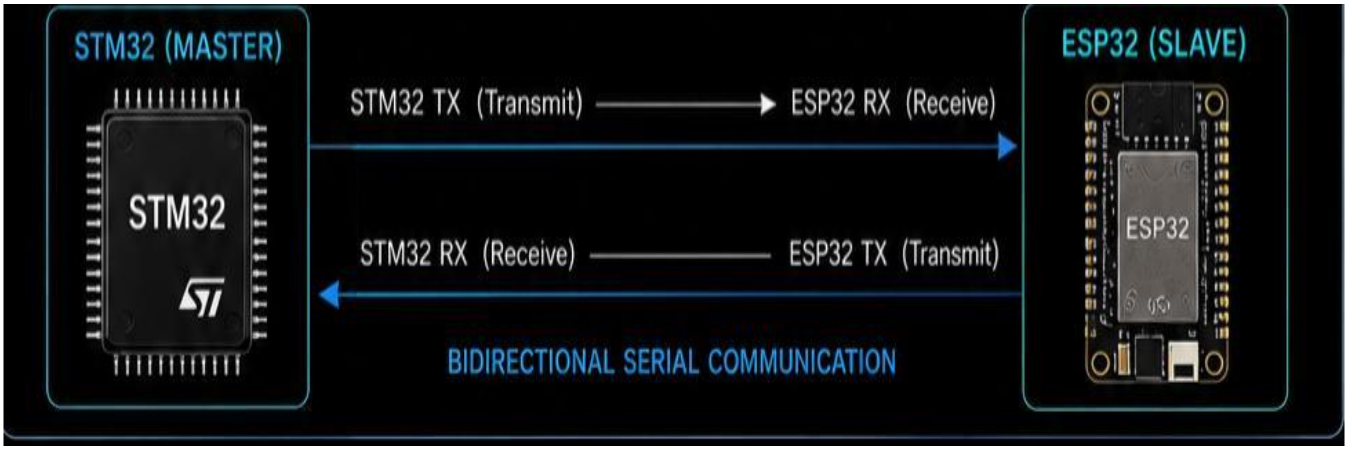
- ➢ **SYSTEM COMMUNICATION ARCHITECTURE**



COMMUNICATION LINKS:

Sensor →STM32	GPIO/ADC/ HX711 Interface: Sensors send raw data to STM32 for acquisition.
STM32→ESP32	UART STM32 transmits processed data packets to ESP32 via UART.
ESP32→ Dashboard	Wi-Fi: ESP32 sends data to the dashboard using Wi-Fi connection.
Dashboard →Database	Internet / API Dashboard stores data to the database using Internet / API.

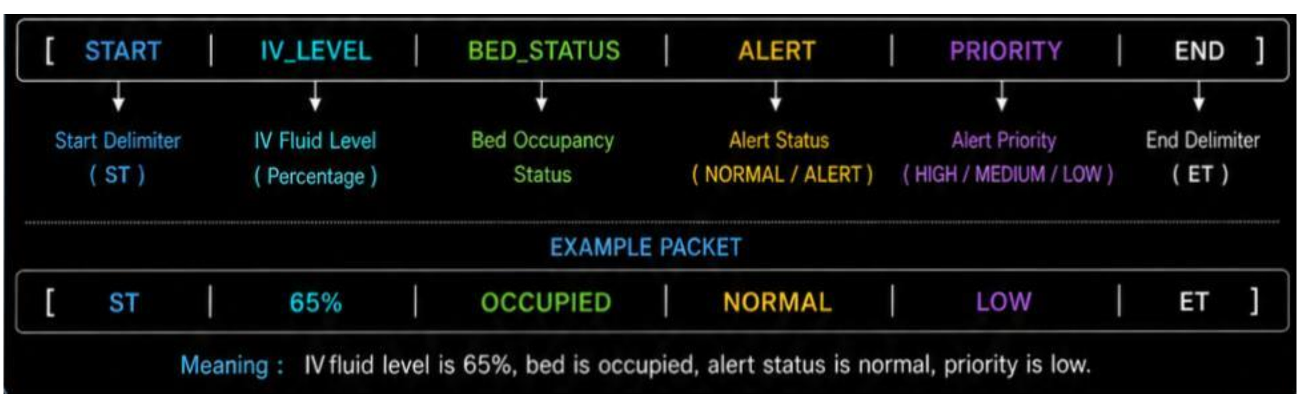
- **UART COMMUNICATION :**



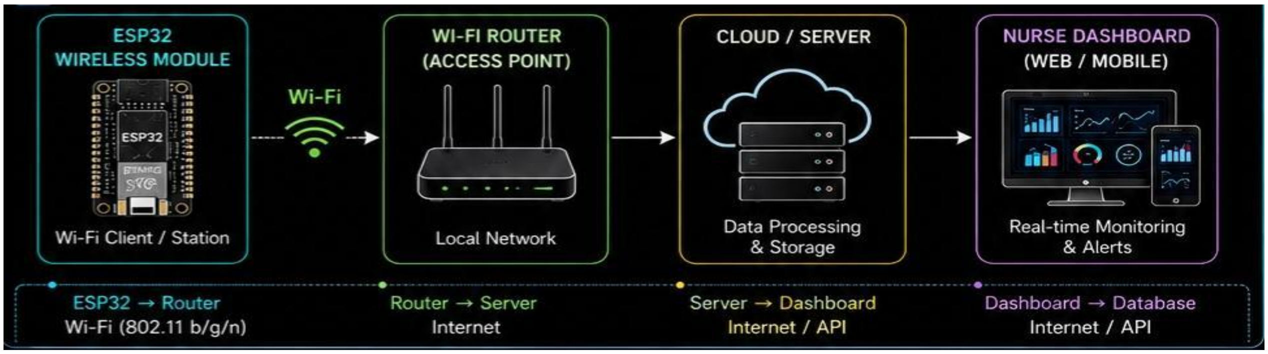
UART SPECIFICATION:

Protocol : UART(Asynchronous serial) Purpose :STM32-ESP32 data transfer Data :IV level,bed status, alert priority Direction :Bidirectional (TX/RX) Baud Rate :115200 bps(configurable) Data Bits :8 Stop Bits :1 Parity : None Flow Control :None Voltage Level :3.3V TTL

UART DATA PACKET FORMAT:



➢ WI-FI COMMUNICATION ARCHITECTURE:



WI-FI COMMUNICATION DETAILS

Protocol	WI-FI (IEEE 802.11 b/g/n)
Purpose	Wireless monitoring and alert transmission
Data	Real-time Sensor status and alerts
Network Mode	Station (Cllent Mode)
Frequency Band	2.4 GHz
Security	WPA2-PSK
Data Rate	Up to 72 Mbps
Topology	Infrastructure Mode
IP Address	DHCP/Static(Configurable)
Transport	TCP/UDP(MQTT/HTTP)

➢ FINAL COMMUNICATION SEQUENCE: